

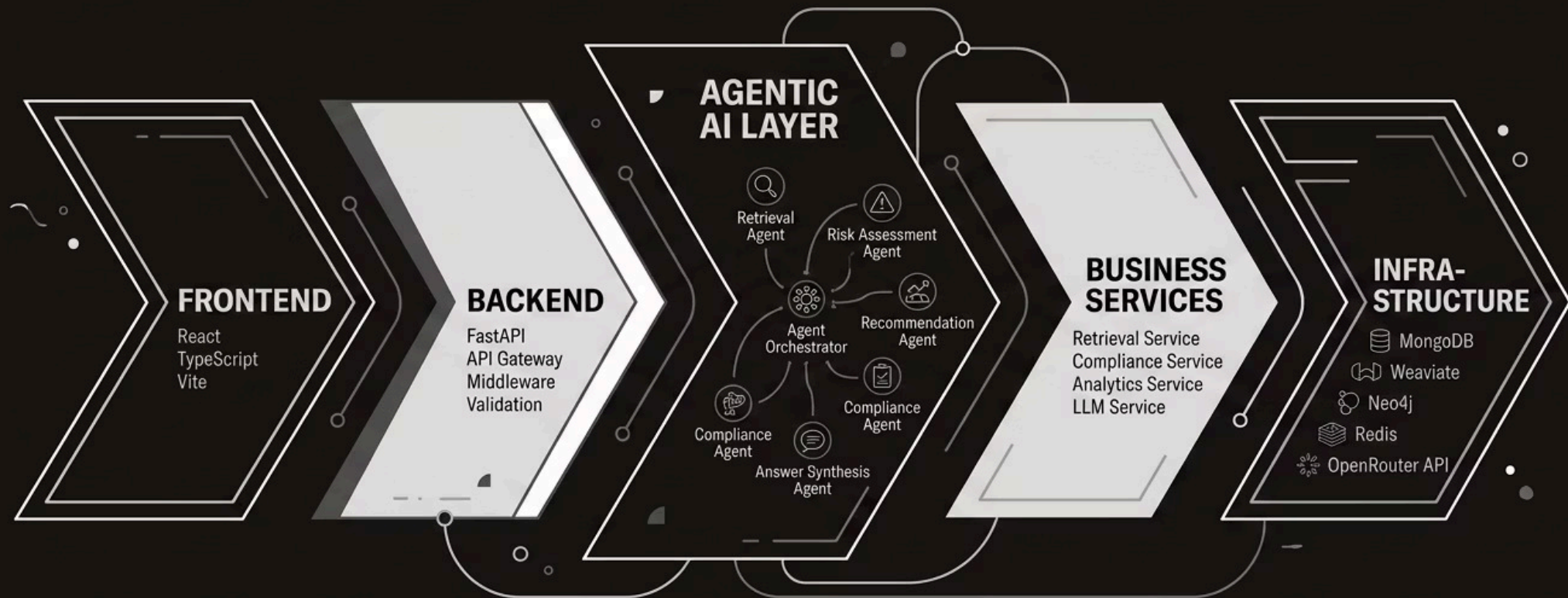


AI-Powered Policy Compliance Intelligence Assistant

Enterprise AI System for Policy Retrieval, Risk Analysis & Compliance
Guidance

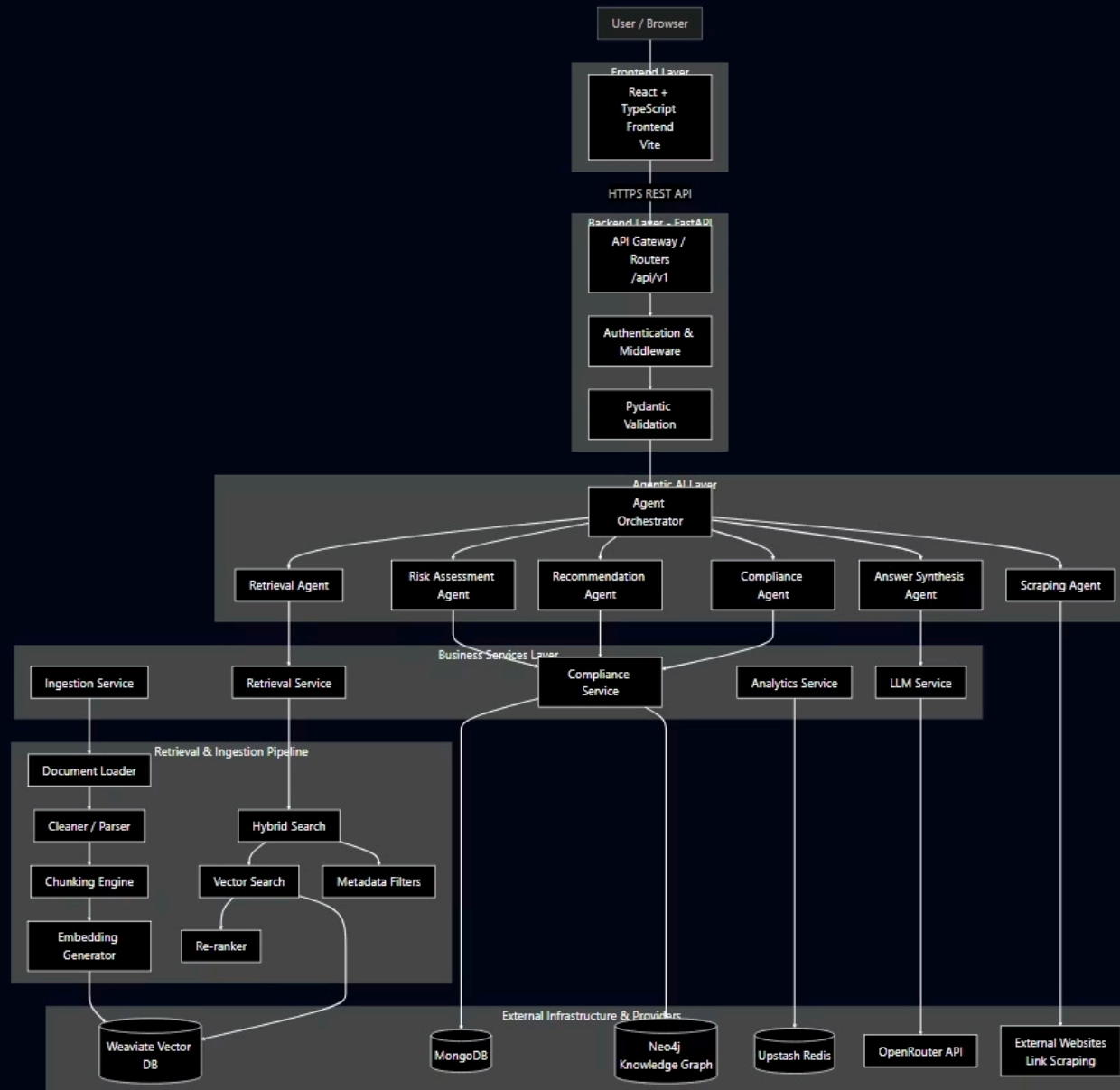
Sayar Chakraborty

High-Level System Architecture



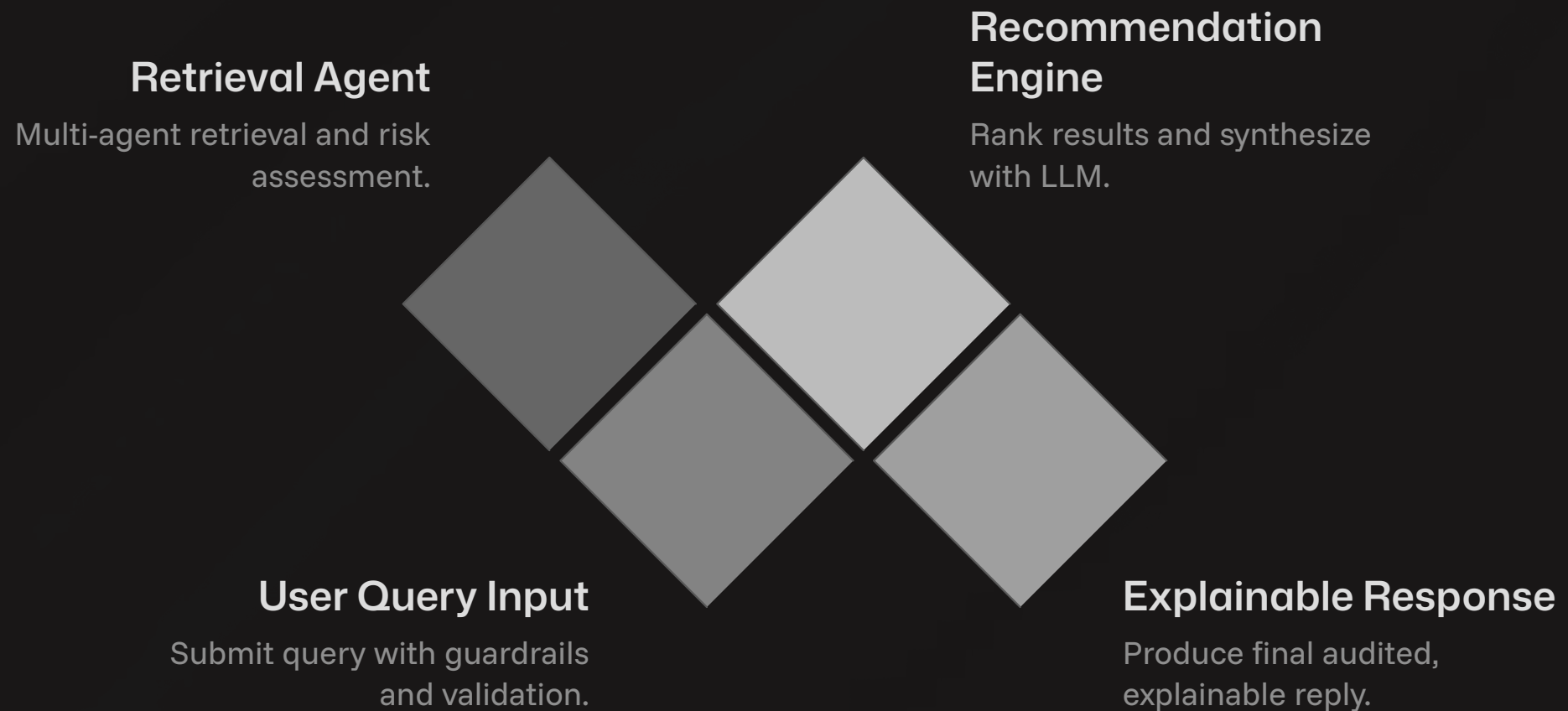
End-to-end data flow from user-facing frontend through agentic AI orchestration to persistent infrastructure — designed for modularity, auditability, and enterprise-scale compliance workloads.

Low Level Architecture Diagram



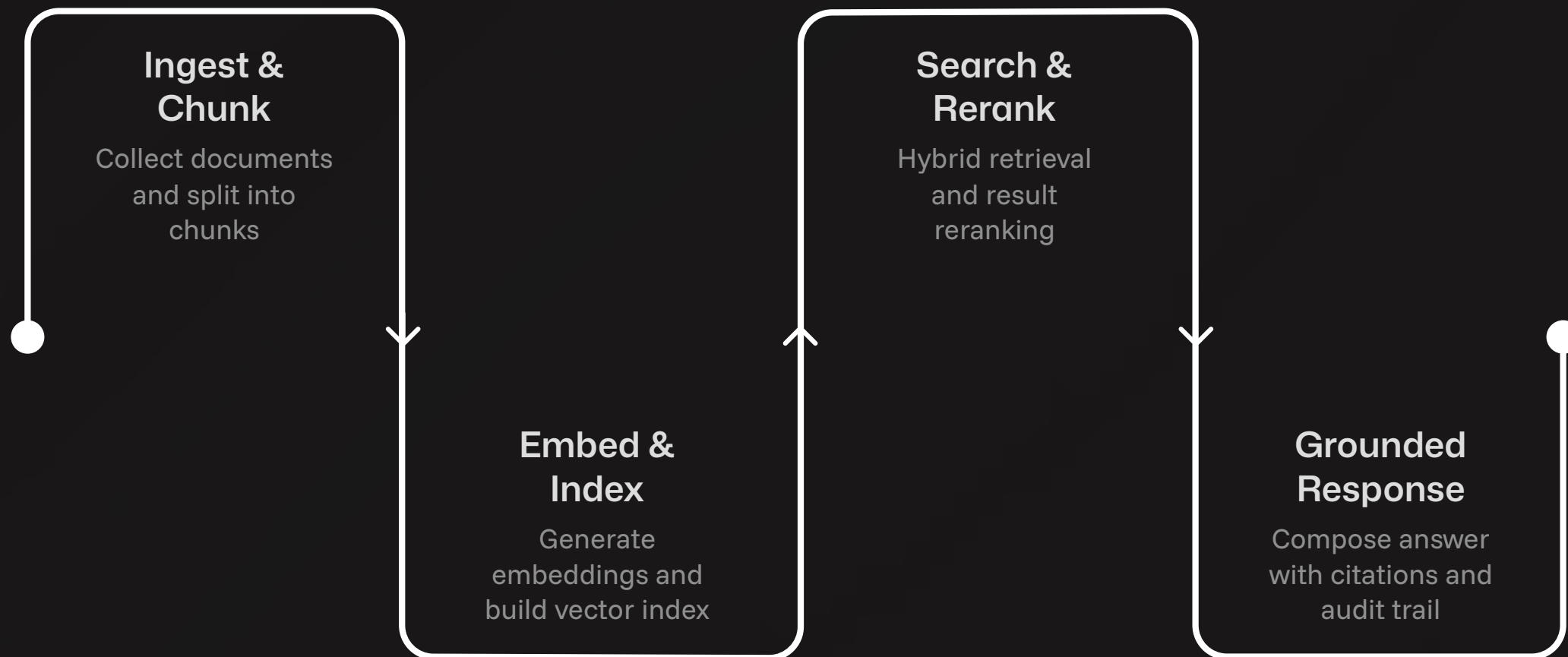
AI Workflow

Agent Orchestration Workflow



Every query passes through structured guardrails, multi-agent retrieval, and risk assessment before a grounded, explainable response is synthesized — ensuring full auditability at each step.

RAG Retrieval Pipeline



Documents are chunked, embedded, and indexed for hybrid semantic-plus-lexical search. Reranking ensures the most relevant evidence is surfaced and cited in every AI-generated response.

Key Engineering Decisions

Hybrid Search

Semantic + lexical retrieval for precision and recall across diverse policy documents

LangGraph Orchestration

Graph-based agent workflows with explicit state management and full traceability

Deterministic Risk Scoring

Rule-based, auditable scoring — not probabilistic black-box inference

Local-First Architecture

On-premise capable with pluggable LLM provider abstraction for flexibility

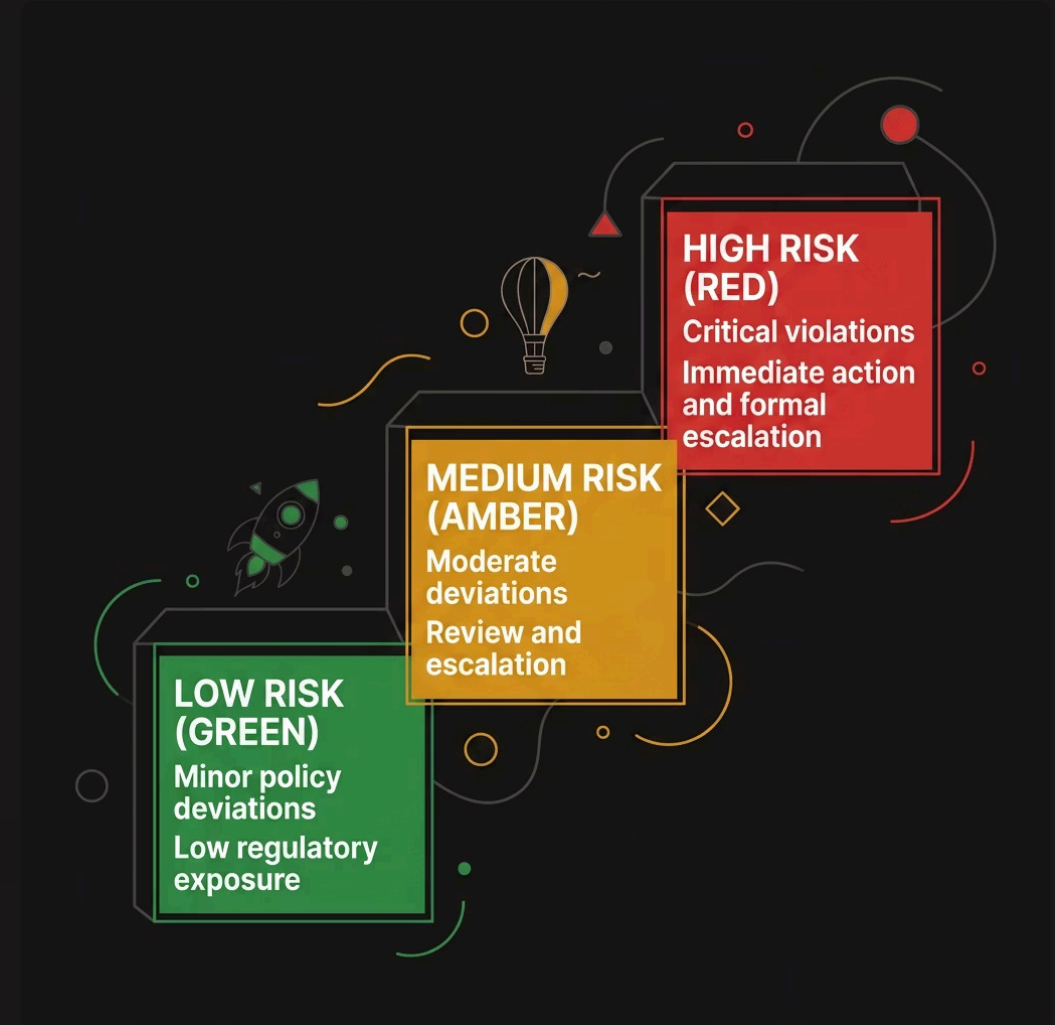
Guardrail Security

Input validation, output filtering at every layer

Risk Scoring & Security Architecture

Compliance Risk Engine

- **Keyword taxonomy** mapped to regulatory clauses
- **Evidence boosting** for high-confidence policy matches
- Three-tier risk levels: **LOW** • **MEDIUM** • **HIGH**
- Fully **explainable scoring** with source citations



Technology Stack

Current Technology Stack



Frontend

React · TypeScript · Vite



Backend

FastAPI · Python



AI / Retrieval

LangGraph · OpenRouter · Weaviate



Databases

MongoDB · Neo4j · Redis

Thank You

